



DESIGN, AUTOMATION
AND TEST IN EUROPE

THE EUROPEAN EVENT FOR
ELECTRONIC SYSTEM DESIGN & TEST

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PALAZZO DELLA GRAN GUARDIA

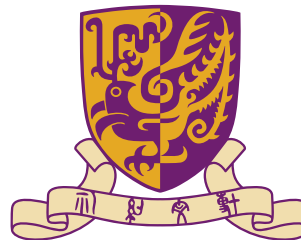


MAEDA: An LLM-Powered Multi-Agent Evaluation Framework for EDA Tool Documentation QA

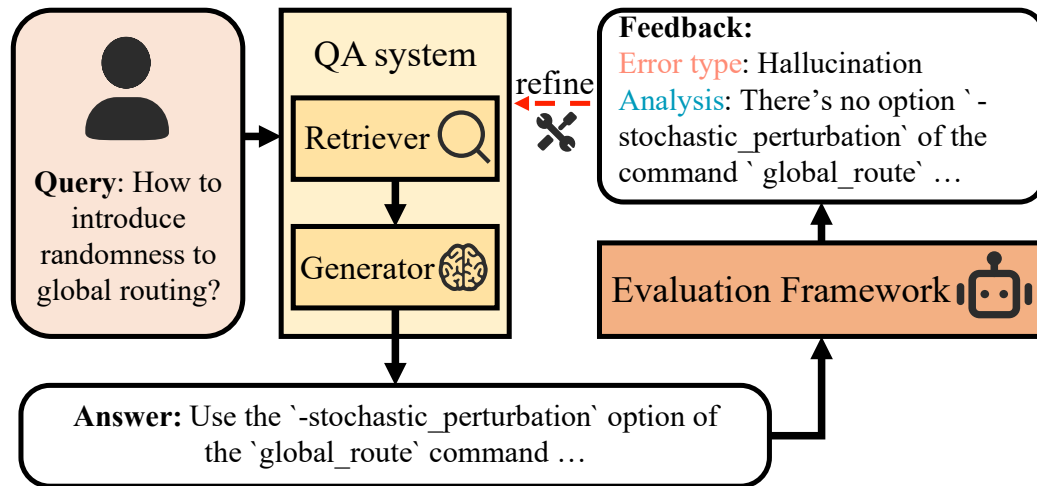
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• Motivation



Existing evaluation **fails** to reveal specific EDA-specific errors in documentation QA systems!

• Problem Defination

Retriever-level Errors

Retrieval Error: This error occurs when retrieved documents lack sufficient evidence to support generated answer.

Inappropriate Refusal:

- False Refusal occurs when the model generates a refusal despite supporting evidence in retrieved documents.
- False Non-Refusal occurs when the model provides an answer to a question that is not answerable based on retrieved documents.

Generator-level Errors

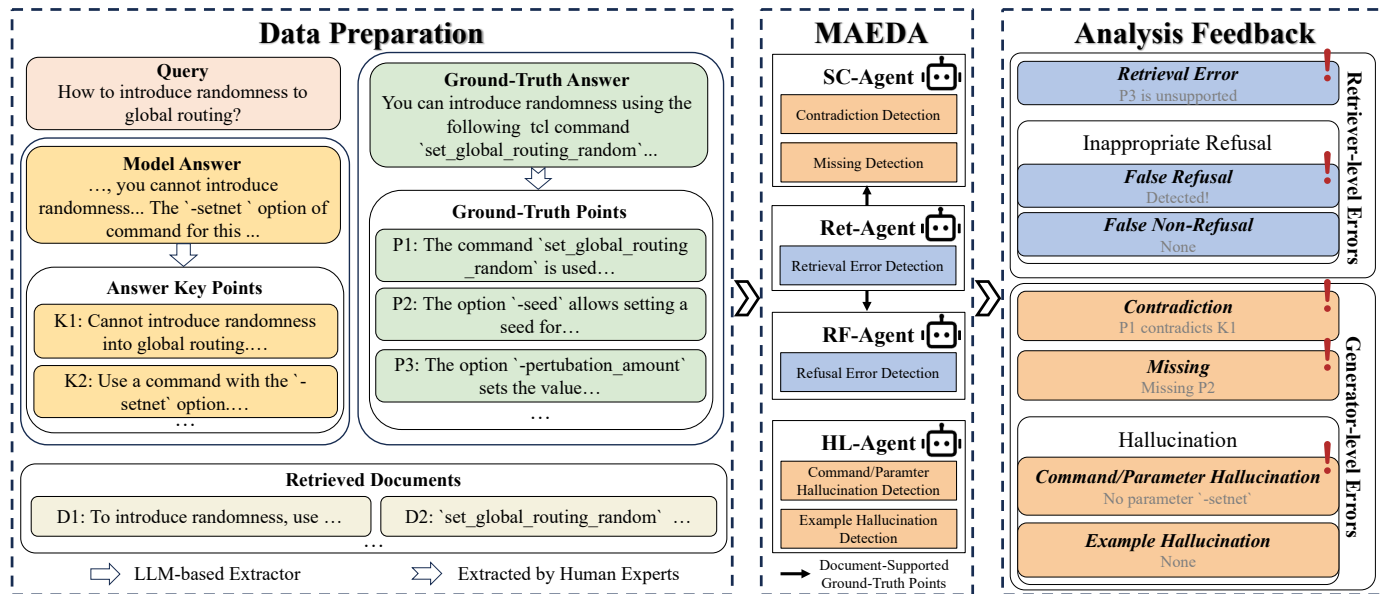
Contradiction: This error occurs when generated answer conflicts with ground-truth.

Missing: This error occurs when ground-truth information is omitted in generated answer.

Hallucination:

- Command/Parameter Hallucination occurs when generated answer fabricates commands or parameters that have no grounding in retrieved documents.
- Example Hallucination occurs when a generated example lacks grounding in retrieved documents.

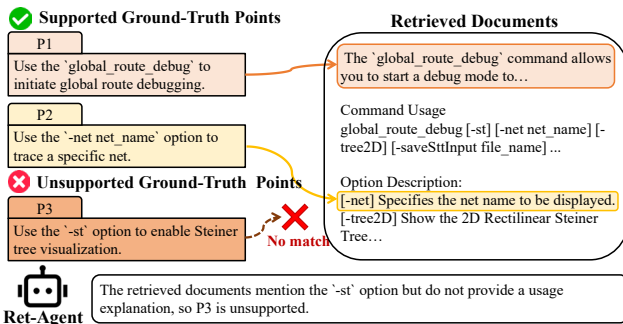
• MAEDA Framework



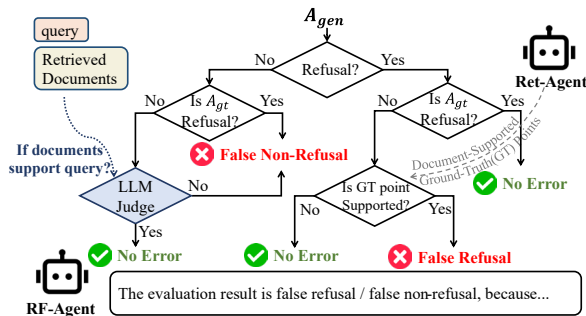
- Point-wise decomposition
- Specialized agents design
- Collaborative multi-agent workflow

• MAEDA Framework

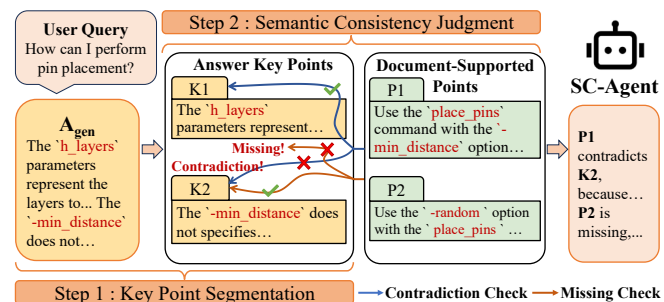
Retrieval Error Identification Agent



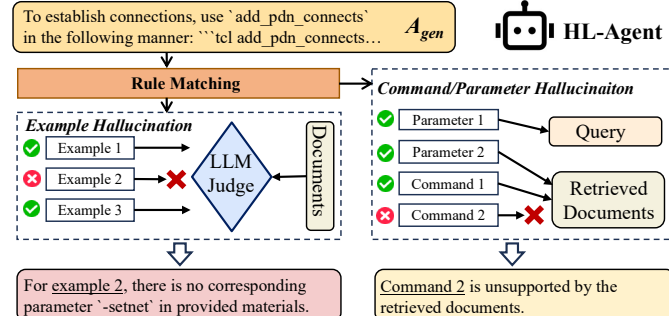
Refusal Evaluation Agent



Semantic Consistency Agent



Hallucination Evaluation Agent



• Experimental Results

TABLE I Model performance on fine-grained error type classification (on OpenROAD Dataset)

Method	Retrieval Error		Contradiction		Missing		Inappropriate Refusal		Hallucination	
	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision
RAGChecker [34]	0.960	0.692	–	–	–	–	–	–	0.486	0.346
RAGEval [35]	0.958	0.926	0.059	0.039	0.825	0.650	–	–	0.095	0.132
G-Eval [37]	0.703	0.703	0.881	0.394	0.772	0.550	0.267	0.143	0.385	0.294
MAEDA w. Qwen3-14B	0.747	0.789	0.902	0.676	0.887	0.601	0.987	0.711	0.935	0.680
MAEDA w. GPT-4o	0.960	0.947	0.922	0.758	0.907	0.633	0.987	0.757	0.860	0.769
MAEDA w. Fine-tuned models*	0.973	0.936	0.941	0.800	0.928	0.726	0.987	0.711	0.935	0.680

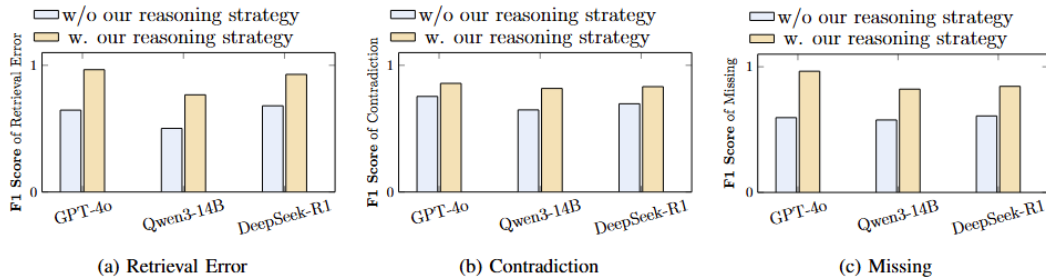
* denotes the use of our fine-tuned models for retrieval error, contradiction and missing, with Qwen3-14B for other error types.

TABLE II Model performance on fine-grained error type classification (on Commercial EDA Tool Dataset)

Method	Retrieval Error		Contradiction		Missing		Inappropriate Refusal		Hallucination	
	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall	Precision
RAGChecker [34]	0.821	0.902	–	–	–	–	–	–	0.648	0.246
RAGEval [35]	0.857	0.528	0.033	0.056	0.788	0.788	–	–	0.038	0.125
G-Eval [37]	0.833	0.641	0.476	0.217	0.852	0.264	0.667	0.08	0.546	0.270
MAEDA w. Qwen3-14B	0.597	0.976	0.667	0.625	0.818	0.600	0.827	0.811	0.872	0.531
MAEDA w. GPT-4o	0.866	0.951	0.767	0.697	0.848	0.800	0.940	0.758	0.782	0.550
MAEDA w. Commercial tool models†	0.896	0.952	0.800	0.727	0.879	0.725	0.827	0.811	0.872	0.531
MAEDA w. Fine-tuned models*	0.866	0.921	0.700	0.700	0.909	0.600	0.827	0.811	0.872	0.531

† denotes the use of similarly trained commercial EDA tool models for retrieval error, contradiction and missing, with Qwen3-14B for other error types.

- Competitive performance with robustness advantages
- Effective hybrid evaluation with strong generalization
- Consistent gains from structured reasoning and alignment



• Conclusion

1. MAEDA: a multi-agent framework for EDA QA evaluation
2. Domain-specific errors with structured reasoning and alignment
3. OpenROAD-grounded benchmark released to support future research.
4. Fine-tuning strategies for domain-specific evaluation
5. Consistent gains over SOTA methods

THANK YOU !